

CSCI 5576

High-Performance Scientific Computing

FORMAT	CATEGORY	CREDITS
On-Campus	Elective	4
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective (On-Campus)	Applied Scientist (83.9 / 100)	64.4 / 100

64

CURRENT AVG SCORE
across 6 job roles

97

POTENTIAL SCORE
with recommended additions

+48

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

Introduces computing systems, software and methods used to solve large-scale problems in science and engineering. Students use high-performance workstations and a supercomputer.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Applied Scientist	83.9	100.0	+16.1	MEDIUM
Machine Learning Engineer	75.1	100.0	+24.9	MEDIUM
AI Engineer	60.8	99.1	+38.3	HIGH
Data Scientist	60.3	100.0	+39.7	HIGH
ML Engineer	54.1	82.0	+27.9	MEDIUM
MLOps Engineer	52.3	100.0	+47.7	HIGH

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (example) appear as gaps across 3 or more job roles.

Model Deployment & Serving 2 of 6 roles

production

Business Communication 1 of 6 roles

business

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

1 **MEDIUM** Consider Model Deployment & Serving content — relevant to 2 industry roles. Could be introduced via case studies or optional readings. Key terms: production.

2 **LOW** Optional: Business Communication — would broaden appeal to one additional role. Candidate terms: business.

GAP KEYWORDS BY ROLE

Applied Scientist

83.9 → 100.0 (+16.1 pts)

discovery scientific research models data learning industry design

Machine Learning Engineer

75.1 → 100.0 (+24.9 pts)

data learning analytics tools solutions business mobile time

AI Engineer

60.8 → 99.1 (+38.3 pts)

data robotics models learning infrastructure build code technology

Data Scientist

60.3 → 100.0 (+39.7 pts)

data characteristics analytics evolution scripting simply social build

ML Engineer

54.1 → 82.0 (+27.9 pts)

media discovery data learning machine models production starting

MLOps Engineer

52.3 → 100.0 (+47.7 pts)

infrastructure data production scientists text learning machine media

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.